

Requirement traceability matrix

Product: PhenoMap

DocID:

Issued by:

Reviewed by:

Appr. by:

Project:

Rev:

Date: _____

Date: _____

Date: _____

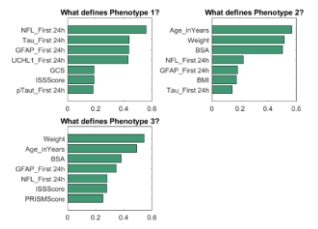
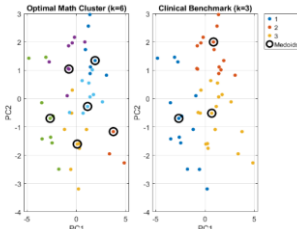
Sign: _____

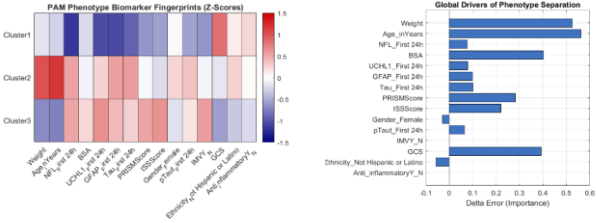
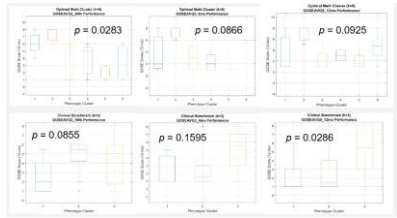
Sign: _____

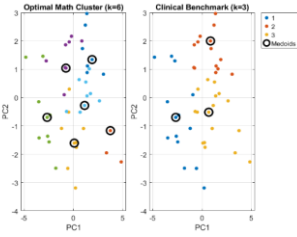
User needs/source of requirements	Req ID	Design input requirements	Design output	Design verification	Acceptance criteria (design input requirements)
As a user, I want the SaMD to show me which patient features are driving each phenotype assignment so that I can understand and trust the algorithm's output.		The SaMD shall display a permutation-based feature importance chart (SHAP-equivalent) in original clinical variable names for each discovered phenotype cluster, generated by a Random Forest classifier trained on all input features.		Confirmed OOB error of 0.089 (accuracy of 91.1%).	As per the requirement
As a user, I want to be able to review each patient's clinical data for each phenotype assignment so that clinician's judgment remains part of the final decision.		The SaMD shall allow an authorized clinician to select any data point to view a patient's clinical fetures, such as ID, physiological markers, biomarker concentrations, etc.		Functional UI testing of the "Data Cursor" or "Point-Click" callback logic within the MATLAB App Designer environment.	As per the requirement
As a user, I want to be able to view the color-coded scatter plot of phenotype cluster assignments for easy comprehension.		The SaMD shall generate a 2-dimensional PCA/UMAP/t-SNE scatter plot where each data point is color-coded by its assigned phenotype.		Visual inspection of the MATLAB figure output generated by the clustering module.	The plot must display distinct color groupings corresponding to the phenotype labels with a visible legend.

As a user, I want to view a side-by-side comparison of clustering output and heatmap or bar plot of top cluster separation drivers.		The GUI shall provide a split-view that displays the cluster plot adjacent to the "Per-Cluster Drivers" bar charts and/or the "Phenotype Signature" heatmap.		Verification of the tiled layout or App Designer UI layout to ensure simultaneous display of clustering and interpretability modules.	The user must be able to view both the patient classification plot and the top drivers heatmap on a single screen without toggling menus.
As a user, I want to be able to compare how distinctly separated are the GOS-E outcome for each cluster at 2-6 weeks, 3-6 months, and 9-12 months.		The SaMD shall generate side-by-side boxplots for GOS-E scores across all clusters at intervals 2-6 weeks, 3-6 months, and 9-12 months.		Statistical analysis of cluster medians and interquartile ranges (IQR) using the Kruskal-Wallis test for each time point.	The plots must be generated successfully in addition to p -values to analyze the significance between clusters distinction.
As a user, I want to be alerted when a patient record exceeds the missing data threshold to ensure reliable phenotype assignment.		The SaMD shall trigger a "High Uncertainty" alert if >30% of required biomarker inputs are missing.		Code review of the missing data threshold logic.	System successfully flags records with >30% missingness.
As a user, I want a reproducible and stable clustering assignment.		The SaMD shall utilize a deterministic PCA- and PAM-based algorithm for dimensionality reduction and cluster assignment to ensure reproducibility.		Re-running the same patient profile through the model multiple times.	Output phenotype must be identical across all iterations.
As a user, I want the clinical data to be preprocessed before analysis.		The SaMD shall perform an log-scale outlier control, missing data imputation, encoding, binary conversion, removal of redundant variables, and Z-score normalization to preprocess the data.		Code review of the preprocessing pipeline.	Processed data must be usable for algorithm use.

As a user, I want a reliable phenotype clustering that is not too overfit nor too underfit.		The SaMD shall utilize and compare a k=3 and k=optimal clustering model to ensure each phenotype maintains a decent sample size.		Review of cluster patient size in the final model output.	Each of the clusters must contain at least 10% of the total patient population.
As a user, I want the system to verify that the patient matches the pediatric population used to train the model to prevent inaccurate or biased phenotype assignments.		The SaMD shall flag outlier patients with skewed data outside of the training cohort.		Outlier detection.	System must detect an outlier warning.

Design validation	Design verification result	Design validation result	Records	Notes
	PASS			
	PASS			
	PASS			

	PASS			
	PASS			
	PASS			
	PASS			<pre> rng(1); % Extract and run PCA X_input = table2array(data_normalized(:, target_vars_final)); [coeff, score, ~, ~, explained] = pca(X_input); % PCA Explain Plot figure('Color','w','Name','PCA Variance Explained'); pareto(explained); % Pareto shows individual and cumulative variance xlabel('Principal Component'); ylabel('Variance Explained (%)'); title('PCA Feature Contribution (Explain Plot)'); grid on; % Select PCs for clustering pca_rank = find(cumsum(explained) > 80, 1); % (cumulative sum > 80%) X_cluster = score(:, 1:pca_rank); fprintf('Retaining %d PCs for HDBSCAN clustering.\n', pca_rank); </pre>
	PASS			%% PREPROCESSING section in code

	PASS			
	PASS			